

Simple Communication

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Abstract

We study multidimensional cheap talk with simple language and aligned preferences. An expert communicates with a decision-maker using a score that aggregates a multidimensional state into a one-dimensional message. Even though the expert and the decision-maker share the same payoffs, the use of simple language introduces strategic frictions. Under quadratic-loss utility, any equilibrium score must be linear in the state or a coarsening of a linear score. As a result, equilibrium payoffs are lower than those achievable under commitment to a score. Finally, for normally distributed states, we characterize the equilibrium linear scores and show they correspond to the ex-ante best and worst linear scores.

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1 Introduction

Decision-makers often seek advice on complex matters. Policymakers consult experts about the impact of new policies; individuals seek professional guidance across diverse domains, including nutritional practices, exercise regimens and financial investments. In principle, experts can write volumes on the potential impacts of a new policy, and dietitians can craft personalized eating plans. In practice, however, advice must be simple. Experts present their findings in short executive summaries. Most people rely on general dietary guidelines, such as the “5 a day” rule, or on standard nutrition labels that cannot be tailored to their specific nutritional needs. In this study, we examine the strategic incentives that arise when advice on complex issues must be simple.

To isolate the effect of simple language, we focus on settings in which the expert and the decision-maker share identical preferences. If the expert could use a language as rich as the object described, revealing every relevant aspect of the object under consideration would be both optimal and an equilibrium strategy. When instead experts communicate using simple language, the nature of communication is not immediately clear.

We explore communication via simple language in a multidimensional cheap talk game. A sender observes a two-dimensional state of the world and sends a cheap-talk message to a decision-maker. The receiver takes a two-dimensional action to minimize a quadratic loss function. Sender and receiver share the same payoffs. The model is also equivalent to one with two receivers, each with a one-dimensional action.

We model simple language by requiring that the sender’s equilibrium strategy maps the two-dimensional state to a one-dimensional message. The lower-dimensional message can be thought of as a simple statistic summarizing the state. A simple message is necessary whenever the receiver lacks the time or capacity to process a rich description of the state, as when a policymaker relies on an executive summary rather than a full technical report. In the multiple receiver interpretation, simple messages are needed whenever the sender can only provide a

single recommendation to all audiences, as in general dietary guidelines for individuals with different health conditions.

Simply requiring one-dimensional messages would not, however, capture a meaningful notion of simple language: since injections from $\mathbb{R}^2$ to $\mathbb{R}$ exist, the sender could fully reveal the state. To prevent injections while imposing natural restrictions on communication, we additionally require that the sender’s equilibrium strategy satisfies a weak form of continuity. Specifically, we focus on mappings from the state space to the real numbers that satisfy a property we dub the *Intermediate Value Property* (IVP). We call such mappings *scores*. The IVP requires that small changes in the state cause at most small changes in the score. The property does not impose monotonicity or other functional-form restrictions and accommodates both continuous and discrete images.¹

We view simplicity as a notion that must hold in equilibrium. In our interpretation, the sender is an expert who, once the state is realized, is free to send any message. Our main result is a characterization of equilibrium scores, that is, scores that constitute the sender’s strategy in a Perfect Bayesian Equilibrium. For any prior distribution, equilibrium scores exist (Proposition 1) and must be either linear in the state or a discrete coarsening of a linear score, up to a relabeling of messages (Proposition 2).

Communicating via linear scores or their coarsening conveys less information than revealing the state. Moreover, there are scores that yield higher payoffs for both the sender and receiver, yet these scores are not equilibrium strategies. The key barrier to effective communication is *strategic frictions*, despite the players’ aligned preferences.

To grasp the source of strategic frictions, recall that we propose an equilibrium-selection criterion rather than a restriction on the strategy space. In other words, once the state is observed, the sender remains free to deviate to any message. A score that yields high payoffs ex-ante might therefore not be an equilibrium score if the sender has a profitable deviation in some states. In Section 3, we illustrate the tension between informativeness and equilibrium behavior in a simplified setting.

A consequence of the linearity of scores is that in equilibrium the two actions are linearly dependent (Proposition 3 and Proposition 4), which in particular implies a monotone relationship between them. If the score represents a dietary guideline, a higher score is either

¹All continuous scores satisfy the IVP, and if the score has a discrete image, e.g., an A-F nutritional label, a marginal change in the state cannot cause the score to jump by more than one grade.

always associated with higher nutritional quality for both healthy consumers and those with a health condition, or always associated with higher quality for one group and lower quality for the other. Simple communication makes it thus impossible to convey non-monotone relations as, for instance, the desirability of salty foods: for some individuals salt is desirable, in moderation, while for others any amount of salt is to be avoided. A second implication of Proposition 3 is that a linear score fully revealing dimension 1 is an equilibrium score only if dimension 2 is mean independent of dimension 1, that is, learning the realization of dimension 1 provides no information about the expected value of dimension 2.

With additional structure on the distribution of states, we can further characterize equilibrium scores. When the state is normally distributed, the best equilibrium score is linear and coincides with the ex-ante optimal linear score (Proposition 5). The only other equilibrium linear score is, by contrast, the ex-ante worst linear score: a score may be state-wise optimal yet ex ante the worst linear one. The key to Proposition 5 is the equivalence between equilibrium scores and the stationary points of the ex-ante payoff maximization problem over linear scores, which correspond to the eigenvectors of a matrix determined by the preference parameters and the correlation structure.

One score positively correlates actions across dimensions, the other negatively correlates them. The optimality of each score depends on the correlation structure: when the dimensions are positively correlated, the ex-ante best linear score correlates the actions and vice versa. Accordingly, an increase in the correlation is good for welfare in the best equilibrium but also increases the cost of coordinating on the wrong equilibrium. While an increase in variance is unambiguously bad in equilibrium, it can have an ambiguous effect on the cost of coordinating on the wrong equilibrium.

1.1 Related Literature

We introduce a notion of equilibrium simple language by requiring the sender to use an aggregator of a multidimensional state, which we dub score. Closest to our paper is the literature studying cheap talk models with aligned preferences and some form of language limitation.² Jäger et al. (2011) consider a sender constrained to use a finite number of messages. They characterize the best equilibria and study their stability. In Blume and Board

²This literature, like us, looks at the consequences of language limitations, not its causes. On the latter topic see Lipman (2003) and Lipman (2025).

(2013) and Blume (2018) uncertainty about the language used can impede communication. These papers model situations in which simplicity is a constraint on the number of messages and therefore on the set of strategies. As a consequence, they find that ex-ante optimal strategies are equilibrium strategies. Similarly, Lipman (2025) uses the fact that optimal strategies are equilibrium strategies to show that there is always an equilibrium in pure strategy when preferences are aligned in cheap-talk games with a possibly constrained set of messages. We take a different approach to modeling simple language: we introduce the notion of scores and require simplicity to be an equilibrium property instead of a constraint on the set of strategies itself. In particular, the optimal scores are not necessarily equilibrium strategies and therefore strategic frictions impose constraints on communication beyond the properties of scores.³

We also relate to the literature on multidimensional cheap talk. This literature has shown that multiple dimensions can be useful for information revelation, e.g., Battaglini (2002), Chakraborty and Harbaugh (2007) and Chakraborty and Harbaugh (2010). In this strand of the literature, the contribution closest to ours is Levy and Razin (2007), who show that correlation across dimensions can limit communication by creating informational spillovers across dimensions. Similar mechanisms are at play in our paper as the sender needs to balance how the score, a one-dimensional object, reveals information across both dimensions.

Finally, there is a strand of the literature in information design, i.e., communication with commitment, where the amount of information transmitted is limited. In Gentzkow and Kamenica (2014), the limitation comes from the cost of designing the experiment, while in Bloedel and Segal (2021) it comes from the information-processing cost faced by the receiver. When considering the commitment problem, we impose a restriction directly on the shape of the information structure, by limiting the sender to select among scores. In this way we are closer to Le Treust and Tomala (2019) and Aybas and Turkel (2024), who consider exogenous constraints on the capacity or cardinality of the message space.

³Other papers consider scores that aggregate a multidimensional variable in different settings. For example, Ball (2025) and Bonatti and Cisternas (2020) study linear scores where sender and receiver have different payoffs and the sender can manipulate the score input.

2 Model

2.1 Setting

Players, States and Strategies. There are two players: a sender and a receiver. The sender has private information about a two-dimensional state of the world, $\theta = (\theta_1, \theta_2) \in \Theta = \mathbb{R}^2$, whose distribution has full support and admits a density function f .⁴ Without loss of generality, we assume that $\mathbb{E}[\theta] = 0$. We also assume that the variance of θ is finite. When there is risk of ambiguity, we write $\tilde{\theta}$ for the random variable and θ for its realization. The receiver takes two actions represented by $a = (a_1, a_2) \in \mathbb{R}^2$. Before the receiver acts, the sender sends a cheap-talk message $m \in \mathcal{M}$ for some measurable space $\mathcal{M} \supseteq \mathbb{R}$.

Payoffs. Sender and receiver share the payoff function

$$u(a, \theta) := -\phi(a_1 - \theta_1)^2 - (a_2 - \theta_2)^2,$$

with $\phi > 0$. Both players want each action to match the corresponding component of the state. The parameter ϕ governs the relative cost of mismatch along each dimension. The sender and receiver's strategies are measurable functions denoted by $\mu : \Theta \rightarrow \mathcal{M}$ and $\alpha : \mathcal{M} \rightarrow \mathbb{R}^2$, respectively. For any $m \in \mathcal{M}$ and $i = 1, 2$, let $\alpha_i(m)$ denote the i -th component of $\alpha(m)$.

Equilibrium Selection. A score s is a non-constant measurable function $s : \Theta \rightarrow \mathbb{R}$ that satisfies the following property:⁵

Intermediate Value Property (IVP): for any $\theta, \theta' \in \Theta$ such that $s(\theta) > s(\theta')$, any $m \in [s(\theta'), s(\theta)] \cap s(\Theta)$ and any continuous function $\gamma : [0, 1] \rightarrow \Theta$ with $\gamma(0) = \theta'$ and $\gamma(1) = \theta$, there is a $t \in [0, 1]$ such that $s(\gamma(t)) = m$.

We denote the collection of all scores by $\mathcal{S}$. Scores aggregate a complex state, represented by $\mathbb{R}^2$, into a simpler message, represented by a real number. The need for aggregation fits at least two scenarios. First, the receiver may lack the time or capacity to process a rich description of the state, as when a policymaker relies on an executive summary rather than

⁴We endow subsets of $\mathbb{R}^n$ with their Euclidean topology and Borel σ -algebra.

⁵Requiring scores to be non-constant simplifies the exposition of our main results at no loss of generality.

a full technical report. Second, a single message may need to serve multiple audiences, as when a public health organization gives public advice that cannot be tailored to individual circumstances.

The Intermediate Value Property requires that along any continuous path through the state space, the score attains every message m used elsewhere ($m \in s(\Theta)$) and lying between the images of the path's endpoints ($m \in [s(\theta'), s(\theta)]$). We discuss scores in more detail in the next section.

We study Perfect Bayesian Equilibria (PBE) in which the sender's strategy takes the form of a score.

Definition 1. *A score s is an equilibrium score if there is a receiver's strategy α such that*

- *for all $\theta \in \Theta$, $m, m' \in s(\Theta)$,*

$$s(\theta) = m \Rightarrow -\phi(\alpha_1(m) - \theta_1)^2 - (\alpha_2(m) - \theta_2)^2 \geq -\phi(\alpha_1(m') - \theta_1)^2 - (\alpha_2(m') - \theta_2)^2,$$

- *for all $m \in s(\Theta)$,*

$$\alpha(m) = \mathbb{E}[\tilde{\theta} | s(\theta) = m].$$

In equilibrium, the sender best-responds to the receiver's strategy, while the receiver best-responds given his beliefs, which are induced by the equilibrium score. We can ignore off-path messages in the definition of equilibrium score as any off-path message can be assigned a belief corresponding to an on-path message.⁶

We call an equilibrium score a *best equilibrium score* if it maximizes the players' payoffs among all equilibrium scores.

We select equilibria in which the sender's strategy is a score. This is different from constraining the sender to use only scores. In our approach, off path the sender can select any strategy. This fits the scenarios described above, where the sender is an expert who can handle the complexity of the state and can, in principle, communicate it in any way. In particular, once the state is realized, the sender has the flexibility to send any message.

⁶The definition implicitly assumes the existence of a conditional expectation. By the Disintegration Theorem (e.g., Kallenberg, 2021, Theorem 3.4), a regular conditional distribution exists and therefore a conditional expectation exists everywhere.

2.2 Discussion of Scores

Scores capture several desirable properties of simple communication: they impose natural constraints, they are flexible, and they prevent injective mappings.

Scores can be continuous or discrete mappings from $\mathbb{R}^2$ to $\mathbb{R}$. When the set $[s(\theta'), s(\theta)] \cap s(\Theta)$ is an interval, the IVP guarantees the familiar Intermediate Value Theorem holds along every curve connecting θ and θ' . When the set $[s(\theta'), s(\theta)] \cap s(\Theta)$ is discrete, the IVP requires that when moving from θ' to θ , the score does not “jump” over any message. As an illustration of the latter case, consider nutritional labels, which map continuous variables — such as grams of sugar and fats — into discrete grades, say A through F. The property ensures that marginal differences in the sugar or fats cannot result in difference of more than one grade. We regard the IVP as a minimal requirement ensuring that the score reflects the underlying state.

Scores can take many forms. All continuous mappings from $\mathbb{R}^2$ to $\mathbb{R}$ are scores, but scores can also be discrete. Examples of scores include:

- $s(\theta) = \beta_1\theta_1 + \beta_2\theta_2$;
- $s(\theta) = \begin{cases} 1 & \text{if } \beta_1\theta_1 + \beta_2\theta_2 \geq c, \\ 0 & \text{otherwise.} \end{cases}$;
- $s(\theta) = \sqrt{(\theta_1 - c_1)^2 + (\theta_2 - c_2)^2}$.

These examples illustrate the flexibility of scores: while the first example is continuous and monotone in each dimension, the second is non-continuous and the third is not monotone in any dimension. The last example measures the distance between the state and a point (c_1, c_2) . If the state represents political positions along two dimensions, this score can be interpreted as a measure of extremism where (c_1, c_2) would be the political center.

Scores prevent injective mappings. By preventing injections from Θ into the message space, scores prevent the sender from fully revealing the state. We see this as a minimal notion of simplicity. Just requiring that the equilibrium strategy is a mapping from Θ to $\mathbb{R}$ would not be enough: injections from $\mathbb{R}^2$ to $\mathbb{R}$ exist. To rule out injections, we additionally require strategies to satisfy the IVP. Our approach parallels the one adopted in the literature on informational requirements in mechanism design. This literature imposes continuity-like assumptions on mechanisms to avoid bijections between sets of different dimensions. See for example, Hurwicz (1972), Mount and Reiter (1974) and Reichelstein (1984).

3 Illustration in a 2x2 Setting

A consequence of our equilibrium selection is that the payoffs from committing to a score can exceed any equilibrium payoffs, despite the common interest between sender and receiver. Before moving to the main analysis, we illustrate this point in a simple 2x2 setting. The analysis in this section is self-contained.

Abusing notation slightly, let $\Theta = \{0, 1\}^2$ and $f(\theta)$ denote the probability of state $\theta \in \{0, 1\}^2$. Our definition of scores does not apply directly to this finite environment but it can be adapted by requiring that the score does not “jump” messages when moving through points that differ by at most one coordinate. Formally, this can be done by replacing the topological notion of path (the function γ in the definition) with the graph-theoretic notion of path.

Viewing the square $\{0, 1\}^2$ as a simple graph where each point is a vertex and the edges of the graph are the edges of the square, we can rewrite the IVP as follows: for any $\theta, \theta' \in \Theta$ with $s(\theta) > s(\theta')$, any $m \in [s(\theta'), s(\theta)] \cap s(\Theta)$ and any path p from θ' to θ , there is some θ'' on the path such that $s(\theta'') = m$.⁷ Figure 1 illustrates four scores in this environment. Dots in the same area represent states to which the score assigns the same message.

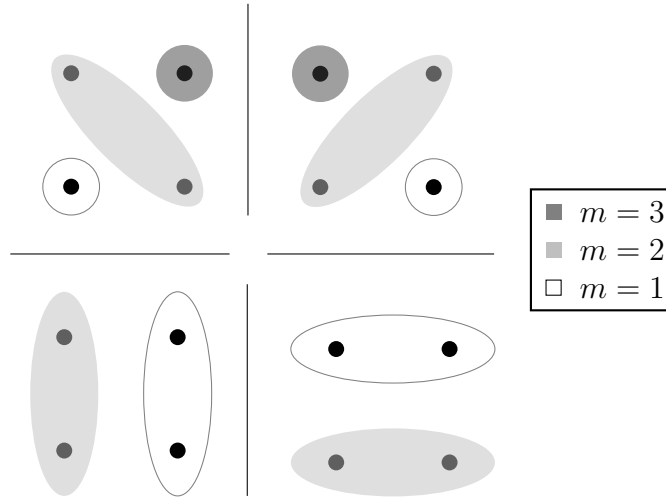


Figure 1: Some scores in $\Theta = \{0, 1\}^2$

We call a score *optimal* if it maximizes the payoffs over all scores, subject to the receiver

⁷See e.g., West (2001) for definitions.

best-replying:

$$\begin{aligned} & \max_{s \in \mathcal{S}} \mathbb{E}[u(\alpha(s(\theta)), \theta)] \\ \text{s.t. } & \alpha(m) = \mathbb{E}[\theta|m], \quad \forall m \in s(\Theta). \end{aligned} \tag{BR}$$

We now show that commitment has value, i.e., it can be the case that none of the optimal scores are equilibrium strategies. Let score s_d be as shown in the top-left panel of Figure 1. The score s_d assigns a common message to states $(0, 1)$ and $(1, 0)$ while assigning unique messages to the other states.

Remark 1. *Score s_d is the unique optimal score if:*⁸

$$\frac{1}{f(0, 0)} + \frac{1}{f(1, 1)} < \frac{1}{f(1, 0)} + \frac{1}{f(0, 1)}, \tag{1}$$

and is not an equilibrium score if:

$$\frac{f(0, 1)}{f(1, 0)} \notin \left[\sqrt{2} - 1, \frac{1}{\sqrt{2} - 1} \right]. \tag{2}$$

There exist probability distributions over states for which both (1) and (2) hold.

The proof of this result and all proofs not in the text are in the appendix. Score s_d is *not* an equilibrium score if the prior probabilities of the two states associated with the same message are too different from one another. The intuition is as follows. Suppose that $\frac{f(0, 1)}{f(1, 0)} > \frac{1}{\sqrt{2} - 1}$, so that the posterior associated with $m = 2$ lies “close” to $(0, 1)$ and “far” from $(1, 0)$. In fact, the posterior is so far from $(1, 0)$ that the score is not an equilibrium strategy: if the receiver expects the sender to communicate according to the score, i.e., $\mu(\theta) = s_d(\theta)$ for all θ , then the sender has a profitable deviation upon observing state $(1, 0)$. Score s_d is shown on the left-hand side of Figure 2. The right-hand side shows the resulting deviating strategy at the ex-ante stage, which involves message $\mu(1, 0) = 3$ instead of $\mu(1, 0) = 2$.

The deviating strategy violates the IVP, as it “jumps” from $\mu(0, 0) = 1$ directly to $\mu(1, 0) = 3$. This strategy is not a score. In general, optimal scores need not be equilibrium strategies precisely because deviations to strategies that are not scores are possible. Relatedly, in some cases, the players are better off if the sender observes only one dimension of the state (see Section G). The intuition is that ignorance reduces the set of potential deviations available

⁸Score s_d is the unique optimal score up to an inconsequential relabeling of the messages.

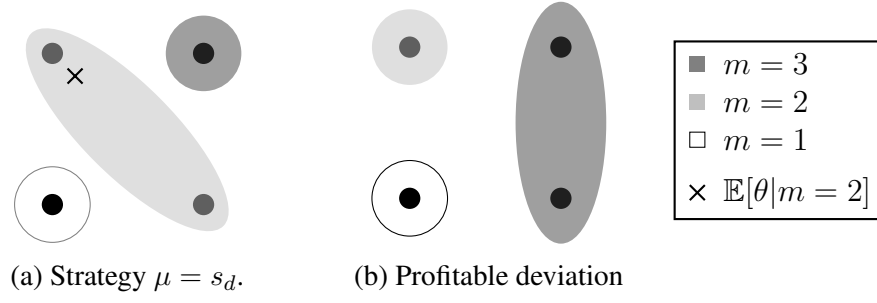


Figure 2

to the sender. This is in contrast with the rest of the literature that studies cheap talk models with aligned preferences (Jäger et al. (2011), Blume and Board (2013) and Blume (2018)) where the constraints on communication bind the message space directly.

4 Analysis

We return here to our model with $\Theta = \mathbb{R}^2$. We present a few preliminary results before turning to the main equilibrium characterization.

4.1 Preliminary Observations: Existence and Commitment

We first establish that an equilibrium score exists.

Proposition 1. *An equilibrium score exists.*

We establish existence of an equilibrium score by showing that there always exists a PBE with two messages in the support of the sender's strategy.⁹ Since any strategy of the sender with two messages trivially satisfies the properties of a score, an equilibrium score exists.

As discussed in Section 2.2, scores cannot map every state into a distinct message.

Remark 2. A score $s : \Theta \rightarrow \mathbb{R}$ is not an injection.

⁹Here we adapt the proof of existence in Jäger et al. (2011) to a potentially unbounded state space.

Proof. If $|s(\Theta)| \leq 2$, then s cannot be injective. Otherwise, take some messages $m, m_1, m_2 \in s(\Theta)$ with $m_1 < m < m_2$, and states $\theta, \theta^1, \theta^2$ such that $s(\theta) = m$, $s(\theta^1) = m_1$ and $s(\theta^2) = m_2$.

We can draw a curve from θ^1 to θ^2 that does not pass through θ . By the IVP, there must be a point θ''' on that curve such that $s(\theta''') = m$. $\square$

Remark 2 implies that scores cannot fully reveal the state. The definition of scores is nonetheless flexible enough to allow *approximate* full revelation. We make this notion precise by considering a sender who can commit to a score, i.e. need not satisfy the incentive-compatibility constraints. We show that such a sender can approximate the payoffs from fully revealing the state. Let v^* denote the *commitment payoffs*, i.e.,

$$v^* := \sup_{s \in \mathcal{S}} \mathbb{E}[u(\alpha(s(\theta)), \theta)] \quad \text{s.t.} \quad \alpha(m) = \mathbb{E}[\theta|m], \quad \forall m \in s(\Theta).$$

Remark 3. The commitment payoffs satisfy $v^* = 0$.

Remark 3 shows that the definition of scores is flexible and in a sense, minimally restricts the mappings from $\mathbb{R}^2$ to $\mathbb{R}$. The result also offers a clean benchmark against which to compare equilibrium scores. It also shows that, if one were interested in the commitment problem of the sender, the IVP does not have enough bite and one should arguably impose stronger restrictions like a form of Lipschitz continuity. Since the commitment problem is not our focus, we allow for the more permissive IVP.

4.2 Main Analysis

In this subsection we show that equilibrium scores must satisfy specific properties. We characterize these properties for arbitrary priors here, and we focus on normally distributed states in the next subsection.

We first introduce two definitions. A score s is *linear* if there exists a vector $\beta \in \mathbb{R}^2 \setminus \{0\}$ such that, for any $\theta \in \Theta$,

$$s(\theta) = \beta_1 \theta_1 + \beta_2 \theta_2.$$

A score s is *coarsely linear* if its image $M \subseteq \mathbb{Z}$ is a set of consecutive integers and there exists a vector $\beta \in \mathbb{R}^2 \setminus \{0\}$ and a countable subset C of $\mathbb{R}$, indexed by M with $c_{m-1} < c_m$,

such that

$$s(\theta) = m \Leftrightarrow \begin{cases} \beta_1\theta_1 + \beta_2\theta_2 \leq c_m & \text{if } m = \min M, \\ c_{m-1} < \beta_1\theta_1 + \beta_2\theta_2 & \text{if } m = \max M, \\ c_{m-1} < \beta_1\theta_1 + \beta_2\theta_2 \leq c_m & \text{otherwise.} \end{cases}$$

Note that in this definition the minimum or the maximum of M may fail to exist. Panel (a) of Figure 3 depicts a linear score: each solid line collects the states that share a common value of $s(\theta)$. Panel (b) illustrates a coarsely linear score: each band collects the states associated with a common value of $s(\theta)$. As the figure suggests, a coarsely linear score is obtained from a linear score by partitioning the image of the latter into a countable collection of intervals.

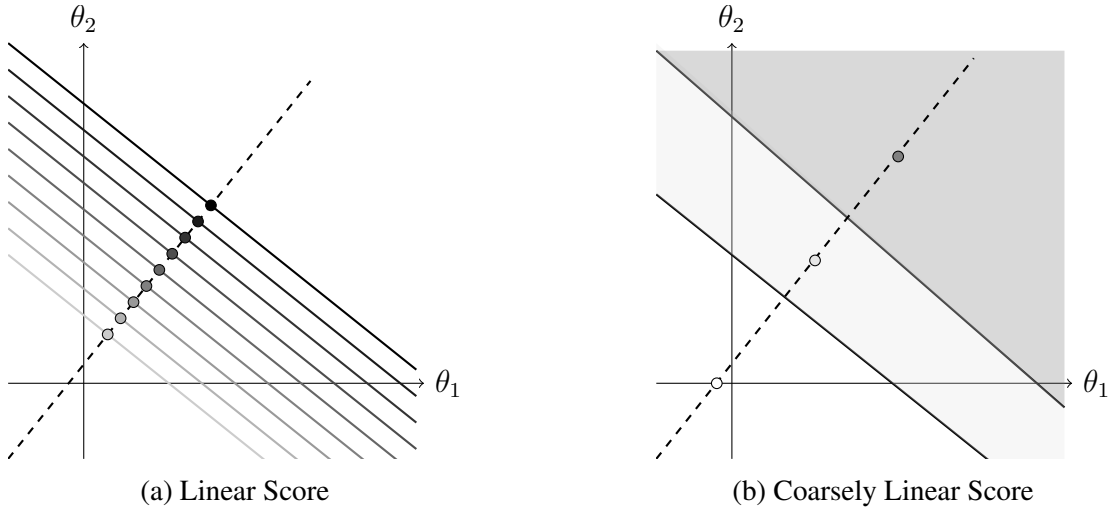


Figure 3: Equilibrium Scores and Actions

We are now ready to state the main result of this section. Two equilibrium scores s and s' are *equivalent* if there are receiver's strategies α, α' that are best-replies to s and s' such that $\alpha(s(\theta)) = \alpha'(s'(\theta))$ for all $\theta \in \Theta$.

Proposition 2. *Any equilibrium score is equivalent to a linear or coarsely linear score.*

As a consequence, all equilibrium payoffs are uniformly below the commitment payoffs.

Proposition 2 records two consequences of strategic frictions in our environment. First, every equilibrium score must satisfy a specific functional-form restriction. Despite the wide range of admissible scores, including some that approach full revelation (Remark 3), in equilibrium, the sender uses a linear or coarsely linear score.¹⁰ Second, despite the alignment

¹⁰Were the loss function not quadratic, the equilibrium forces would induce alternative constraints on the

of preferences between sender and receiver, equilibrium payoffs are lower than commitment payoffs.¹¹ We quantify the loss from strategic frictions in Section 4.3, where we assume normally distributed states.

To see the logic behind the first half of Proposition 2, observe that, given a belief about the sender's strategy, the receiver takes the action $\alpha(m) = \mathbb{E}[\theta|m]$. The sender problem in state θ , given this belief, is to choose the message m' that minimizes the loss function:

$$\min_{m'} (\phi(\theta_1 - \alpha_1(m'))^2 + (\theta_2 - \alpha_2(m'))^2).$$

Because the sender minimizes a weighted Euclidean distance, the set of states indifferent between any two messages must be a line. The IVP requires that these indifference lines do not cross. Together these observations imply that any equilibrium score with a discrete image must be coarsely linear, with linear scores arising as a limiting case. The remainder of the proof shows that when the image of the score is not discrete, linear scores are the *only* scores compatible with the equilibrium conditions.

The functional-form restriction in Proposition 2 also gives some insight into the actions that the receiver takes in equilibrium. We begin with linear scores. Let s_β denote the linear score determined by β .

Proposition 3. *Let $\beta \in \mathbb{R}^2 \setminus \{0\}$. The following are equivalent:*

1. *The linear score s_β is an equilibrium score.*
2. *For every m ,*

$$\mathbb{E}[\tilde{\theta} | s_\beta(\theta) = m] = \left(\frac{\beta_1}{\beta_1^2 + \phi\beta_2^2} m, \frac{\phi\beta_2}{\beta_1^2 + \phi\beta_2^2} m \right). \quad (3)$$

In equilibrium the receiver, upon observing message m , takes the action $\alpha(m) = \mathbb{E}[\theta|m]$. The proposition establishes that a linear score is an equilibrium score if and only if the set of actions induced by the score s forms the line described in (3). For each m , the functional form in (3) guarantees incentive compatibility for every type on the line $s_\beta^{-1}(m)$. Since the

admissible functional forms of the score. In light of this observation, Proposition 2 should not be interpreted as demonstrating that linear strategies are inherently special, but rather as illustrating that strategic frictions generate functional-form restrictions on the structure of communication.

¹¹This result is stronger than the one in the 2x2 setting, where for certain distribution of states, equilibrium payoffs are as high as commitment ones.

sender minimizes a weighted Euclidean distance, the incentive-compatibility conditions are equivalent to an orthogonality condition with $s_\beta^{-1}(m)$, where orthogonality is defined with respect to an inner product induced by the payoff function. When $\phi = 1$, the set of induced actions must be a line perpendicular to $s_\beta^{-1}(m)$ for every m . In Panel (a) of Figure 3, the dashed line represents the set of equilibrium actions for an equilibrium linear score.

Two observations follow. First, under any equilibrium linear score, the induced mapping from messages to each action of the receiver is monotone in each dimension.¹² Consequently, in equilibrium the two actions are either substitutes or complements. When both $\beta_1, \beta_2 > 0$, a higher message is associated with a higher state in both dimensions: $\mathbb{E}[\theta_i|m]$ is increasing in m for $i = 1, 2$. If, for instance, the score rates a movie by considering its aesthetic quality, θ_1 , and entertainment value, θ_2 , then a higher rating indicates that the movie has a higher expected quality along both dimensions. If instead β_1 and β_2 have opposite signs, the score may be interpreted as a relative measure: a higher message is associated with a higher expected value in one dimension and lower expected value in the other.

Second, a linear score that fully reveals one dimension, say θ_1 , is an equilibrium score if and only if the conditional expectation of θ_2 given θ_1 , $\mathbb{E}[\theta_2|\theta_1]$, is independent of the realization of θ_1 . The intuition is that if θ_1 conveys information about the expected value of θ_2 , the sender has an incentive to deviate from revealing θ_1 in order to correct the receiver's expectations about θ_2 . This information spillover echoes the finding in Levy and Razin (2007) who show that misalignment in one dimension can hinder communication in another dimension where receiver and sender have aligned preferences.

We next present a counterpart of Proposition 3 for coarsely linear scores. For any $\beta \in \mathbb{R}^2$ and countable $C \subset \mathbb{R}$, let $s_{\beta,C}$ denote the coarsely linear score defined by β and C .

Proposition 4. *Let $\beta \in \mathbb{R}^2 \setminus \{0\}$ and let $C \subset \mathbb{R}$ be a countable set.*

The score $s_{\beta,C}$ is an equilibrium score

$\Rightarrow$ *There exists $b_1, b_2 \in \mathbb{R}$ such that for every m ,*

$$\mathbb{E}[\tilde{\theta}|s_{\beta,C}(\theta) = m] \in \left\{ \left(\frac{\beta_1}{\beta_1^2 + \phi\beta_2^2} m' + b_1, \frac{\phi\beta_2}{\beta_1^2 + \phi\beta_2^2} m' + b_2 \right) : m' \in \mathbb{R} \right\}.$$

For coarsely linear scores, as for linear ones, the conditional expectations, and therefore the

¹²Note that in general linear scores do not necessarily induce a monotone mapping from messages to actions.

actions taken in equilibrium, must lie on a line determined by β . Panel (b) of Figure 3 shows the equilibrium actions for an equilibrium coarsely linear score. An “if and only if” characterization analogous to Proposition 3, would require, in addition, that the conditional expectations lie at points such that whenever θ satisfies $\beta_1\theta_1 + \beta_2\theta_2 = c_m$, the type θ is indifferent between messages m and $m + 1$.

4.3 Normally Distributed State

We now impose more structure on the prior by assuming that the state is normally distributed, and we fully characterize the linear equilibrium scores, including their welfare properties. The characterization of linear scores enables us to determine the best equilibrium score. The results remain valid under elliptical distributions, a broader class of distributions that includes the multivariate normal, logistic, and t distributions.

Let $\mathcal{S}_l := \{s : \Theta \rightarrow \mathbb{R} : s \text{ is linear}\}$. We refer to a score as an *ex-ante best linear score* if it solves the problem:

$$\max_{s \in \mathcal{S}_l} \mathbb{E}[-\phi(\alpha_1(s(\theta)) - \theta_1)^2 - (\alpha_2(s(\theta)) - \theta_2)^2] \quad \text{s.t. } \alpha(m) = \mathbb{E}[\theta|m], \quad \forall m \in s(\Theta).$$

Analogously, we refer to a score as an *ex-ante worst linear score* if it solves

$$\min_{s \in \mathcal{S}_l} \mathbb{E}[-\phi(\alpha_1(s(\theta)) - \theta_1)^2 - (\alpha_2(s(\theta)) - \theta_2)^2] \quad \text{s.t. } \alpha(m) = \mathbb{E}[\theta|m], \quad \forall m \in s(\Theta).$$

Let

$$\Sigma := \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

be a covariance matrix and

$$\Phi := \begin{pmatrix} \phi & 0 \\ 0 & 1 \end{pmatrix}.$$

From now on, we identify a linear score $s_\beta(\theta) = \beta'\theta$ with the weights $\beta = (\beta_1, \beta_2)'$.¹³

Proposition 5. *Let $\theta \sim N(0, \Sigma)$. The equilibrium linear scores are the eigenvectors of $\Phi\Sigma$. These are the ex-ante best and worst linear scores.*

Corollary 1. *Let $\theta \sim N(0, \Sigma)$. An ex-ante best linear score is a best equilibrium score.*

¹³We use the convention that when writing a vector as a matrix, it is a column vector.

Proposition 5 shows that when the state is normally distributed, the best *linear* score is achievable in equilibrium. A second linear equilibrium also exists, however, corresponding to the worst possible linear score. The key idea underlying the proof is the following. A linear score β , with corresponding strategy α , is an equilibrium score if the indifference curve of each type θ sending message m , $\{a \in \mathbb{R}^2 : u(a, \theta) = u(\alpha(m), \theta)\}$, is tangent to the curve $\{\alpha(m) : m \in \mathbb{R}\}$. We show that the linear scores β satisfying these tangency conditions are the eigenvectors of $\Phi\Sigma$. These eigenvectors, in turn, solve the first-order conditions of the ex-ante maximization problem.¹⁴

Corollary 1 follows directly from the characterization of Proposition 5. Any coarsely linear score with weights β is dominated by the linear score with the same weights β . Therefore, an ex-ante best linear score dominates all linear and coarsely linear scores. Since an ex-ante best linear score is itself an equilibrium score, it is also the best equilibrium score.

We can compute the equilibrium linear scores explicitly. Note that for any constant $c \neq 0$, two linear scores β' and β'' such that $\beta' = c\beta''$ induce the same distribution over actions. Hence, any linear score is determined by the ratio β_1/β_2 , when this ratio exists.

Corollary 2. *Suppose $\rho \neq 0$. The best equilibrium linear scores, $\beta^+ = (\beta_1^+, \beta_2^+)$, and the worst equilibrium linear scores, $\beta^- = (\beta_1^-, \beta_2^-)$, are determined by the ratios*

$$\frac{\beta_1^+}{\beta_2^+} = \frac{\phi\sigma_1^2 - \sigma_2^2 + \sqrt{(\phi\sigma_1^2 - \sigma_2^2)^2 + 4\phi\rho^2\sigma_1^2\sigma_2^2}}{2\rho\sigma_1\sigma_2} \geq 0 \Leftrightarrow \rho > 0,$$

$$\frac{\beta_1^-}{\beta_2^-} = \frac{\phi\sigma_1^2 - \sigma_2^2 - \sqrt{(\phi\sigma_1^2 - \sigma_2^2)^2 + 4\phi\rho^2\sigma_1^2\sigma_2^2}}{2\rho\sigma_1\sigma_2} \leq 0 \Leftrightarrow \rho > 0.$$

If $\rho = 0$ and $\phi\sigma_1^2 \neq \sigma_2^2$, then in each equilibrium score either $\beta_2 = 0$ or $\beta_1 = 0$, so the equilibrium scores fully reveal one dimension. Moreover,

$$\beta_2^+ = 0 \Leftrightarrow \phi\sigma_1^2 > \sigma_2^2,$$

$$\beta_2^- = 0 \Leftrightarrow \phi\sigma_1^2 < \sigma_2^2.$$

The proofs of Corollary 2 and Corollary 3 below follow directly from computing the eigenvalues and eigenvectors of $\Phi\Sigma$ and are omitted. When $\rho = 0$ and $\phi\sigma_1^2 = \sigma_2^2$, all linear scores

¹⁴The proof of Proposition 5 is general and can be extended to arbitrary dimensions of the state and action space and the score is from $\mathbb{R}^n$ to $\mathbb{R}$. When the dimension is larger than two, the set of equilibrium linear scores coincides with the set of stationary points of the ex-ante maximization problem.

are equilibrium scores.

When the correlation between the two dimensions is positive ($\rho > 0$), the ex-ante best linear score satisfies $\beta_1/\beta_2 > 0$, inducing positively correlated actions by the receiver. When the correlation is negative, the ex-ante best linear score satisfies $\beta_1/\beta_2 < 0$, while the ex-ante worst satisfies $\beta_1/\beta_2 > 0$. The worst score can be a natural candidate for an equilibrium score. For example, if movie critics use a rating system in which a higher rating indicates higher aesthetic or entertainment value but these two dimensions are negatively correlated, then the equilibrium score has poor welfare properties.

Finally, as in Proposition 3, revealing only one dimension is an equilibrium only if the two dimensions are uncorrelated. In this case, the best linear score depends on the loss from mismatch ϕ and the variance of each dimension.

We conclude with a comparative statics exercise on the payoffs associated with the linear equilibria.

Corollary 3. *The payoffs associated with the best and worst linear scores are, respectively:*

$$\begin{aligned} v^+ &:= \frac{-\phi\sigma_1^2 - \sigma_2^2 + \sqrt{(\phi\sigma_1^2 - \sigma_2^2)^2 + 4\phi\rho^2\sigma_1^2\sigma_2^2}}{2}, \\ v^- &:= \frac{-\phi\sigma_1^2 - \sigma_2^2 - \sqrt{(\phi\sigma_1^2 - \sigma_2^2)^2 + 4\phi\rho^2\sigma_1^2\sigma_2^2}}{2}. \end{aligned}$$

Hence,

1. $v^+(\rho)$ is increasing in $|\rho|$ and $v^-(\rho)$ is decreasing in $|\rho|$.
2. $v^+(\sigma_1^2, \sigma_2^2)$ and $v^-(\sigma_1^2, \sigma_2^2)$ are decreasing in (σ_1^2, σ_2^2) .
3. $v^+(\sigma_1^2) - v^-(\sigma_1^2)$ is convex in σ_1^2 , and non-monotone for $|\rho| < \frac{\sqrt{2}}{2}$.

The first point shows that an increase in correlation is good for welfare in the best equilibrium but also raises the cost of coordinating on the wrong equilibrium. Intuitively, higher correlation brings the two-dimensional state space closer to a one-dimensional one. In this case, the best linear score captures the state well, but the worst score pools highly probable states together.

An increase in variance is unambiguously bad in any equilibrium. As the third point shows, however, it can have an ambiguous effect on the cost of coordinating on the wrong equilib-

rium. For low values of σ_1^2 , increasing the variance decreases the gap between v^+ and v^- , provided the correlation is low enough. For high values of σ_1^2 , the gap always increases.

5 Conclusion

We study a cheap-talk game with aligned preferences in which the sender is required to use a score in equilibrium. We show that this requirement introduces strategic frictions despite the alignment of preferences between sender and receiver. These frictions drive a wedge between the equilibrium and the commitment payoffs, and they also impose structure on the functional form of equilibrium scores.

The multidimensionality of our model plays a central role. Indeed, if the state were one-dimensional, then, holding fixed the quadratic-loss payoffs, a best equilibrium score would yield the commitment payoffs. In a one-dimensional setting, a score can be defined in several ways. Let $\Theta \subseteq \mathbb{R}$ and let the sender send messages in $M \subseteq \mathbb{R}$. A score is a function s satisfying

1. $s : \Theta \rightarrow M$ and
2. s satisfies the IVP.

If $M = \mathbb{R}$, full revelation is possible, so the best equilibrium yields the commitment payoffs. If $M = \{1, \dots, n\}$, the equivalence between best equilibrium and commitment payoffs follows from the observation that for any given score and its associated belief, the most profitable deviation is itself a score. Hence, if such a deviation were profitable, the original score could not have attained the commitment payoffs in the first place. This is the key difference with the two-dimensional case, in which a profitable deviation may be a strategy that is not a score.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude for language editing. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

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A Proof of Remark 1

Let s_D , s_v and s_h denote, respectively, the scores depicted in the top-right, bottom-left and bottom-right panel of Figure 1. It is easy to verify that any optimal score must be identical, up to a relabeling of messages, to a score in the set $\{s_d, s_D, s_h, s_v\}$. Let the payoffs associated with s_d , s_D , s_h and s_v be respectively, u_d , u_D , u_h and u_v so that:

$$\begin{aligned} u_D &:= -2g(f(0,0), f(1,1)); \\ u_d &:= -2g(f(1,0), f(0,1)); \\ u_v &:= -g(f(0,0), f(0,1)) - g(f(1,0), f(1,1)); \\ u_h &:= -g(f(0,0), f(1,0)) - g(f(0,1), f(1,1)), \end{aligned}$$

where $g(x, y) := \frac{xy}{x+y}$.

Condition (1) is equivalent to $g(f(1,0), f(0,1)) < g(f(0,0), f(1,1))$, i.e., $u_d > u_D$.

Let $a := f(0,0)$, $b := f(1,1)$, $c := f(1,0)$ and $d := f(0,1)$. Condition (1) can be written as:

$$\frac{1}{a} + \frac{1}{b} < \frac{1}{c} + \frac{1}{d}. \quad (\text{A.1})$$

We show that (A.1) implies

$$\frac{ac}{a+c} + \frac{bd}{b+d} > \frac{2cd}{c+d}. \quad (\text{A.2})$$

For any positive reals p and q , the Arithmetic Mean-Harmonic Mean inequality gives

$$\frac{1}{p} + \frac{1}{q} \geq \frac{4}{p+q}.$$

Set $p = \frac{1}{a} + \frac{1}{c}$ and $q = \frac{1}{b} + \frac{1}{d}$. Then

$$\frac{ac}{a+c} + \frac{bd}{b+d} = \frac{1}{p} + \frac{1}{q} \geq \frac{4}{p+q} = \frac{4}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}. \quad (\text{A.3})$$

From (A.1),

$$\frac{4}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}} > \frac{4}{2\left(\frac{1}{c} + \frac{1}{d}\right)} = \frac{2}{\frac{1}{c} + \frac{1}{d}} = \frac{2cd}{c+d}. \quad (\text{A.4})$$

Chaining (A.3) and (A.4) we get (A.2). Condition (A.2) in turn is equivalent to $u_d > u_h$.

The same argument, can be used to show (A.1) implies

$$\frac{ad}{a+d} + \frac{bc}{b+c} > \frac{2cd}{c+d}, \quad (\text{A.5})$$

which is equivalent to $u_d > u_v$. We conclude that s_d is the unique optimal score.

Consider now a PBE such that $\mu(\theta) = s_d(\theta)$ for all θ . In such a PBE, $\mu(0, 0) = 1$, $\mu(0, 1) = \mu(1, 0) = 2$ and $\mu(1, 1) = 3$; $\alpha(1) = (0, 0)$, $\alpha(2) = (\frac{f(1,0)}{f(1,0)+f(0,1)}, \frac{f(0,1)}{f(1,0)+f(0,1)})$ and $\alpha(3) = (1, 1)$. Note that $u(\alpha(3), (1, 0)) = u(\alpha(1), (1, 0)) = -1$ hence

$$u(\alpha(2), (1, 0)) \geq u(\alpha(1), (1, 0)) \Leftrightarrow u(\alpha(2), (1, 0)) \geq u(\alpha(3), (1, 0)) \Leftrightarrow \frac{f(1,0)}{f(0,1)} \geq \sqrt{2} - 1,$$

while

$$u(\alpha(2), (0, 1)) \geq u(\alpha(1), (0, 1)) \Leftrightarrow u(\alpha(2), (0, 1)) \geq u(\alpha(3), (0, 1)) \Leftrightarrow \frac{f(1,0)}{f(0,1)} \leq \frac{1}{\sqrt{2} - 1}.$$

A necessary condition for s_d to be an equilibrium score is therefore that

$$\frac{f(1,0)}{f(0,1)} \in \left[\sqrt{2} - 1, \frac{1}{\sqrt{2} - 1} \right],$$

hence when condition 2 holds, s_d is not an equilibrium score.

To conclude the proof we need to show that conditions (1) and (2) can be satisfied at the same time. This is immediate. Pick some $f(0, 1)$ and $f(1, 0)$ that satisfy (2). Let $f(0, 0) = \epsilon$. Any value of $\epsilon \in (0, 1 - f(0, 1) - f(1, 0))$ is admissible. For ϵ close enough to 0, condition (1) holds.

B Proof of Proposition 1

We first establish that there always exists an equilibrium with two messages.

Lemma A1. *There exists a Perfect Bayesian Equilibrium in which the sender chooses a strategy $\mu : \Theta \rightarrow \{1, 2\}$.*

Proof. As a first step, we establish that the function

$$v(a^I, a^{II}) := \int_{\Theta} \max\{u(a^I, \theta), u(a^{II}, \theta)\} dF$$

is continuous. To show this, we apply the Dominated Convergence Theorem.

Take two converging sequences in $\mathbb{R}^2$, $(a^{I,n}, a^{II,n}) \rightarrow (a^I, a^{II})$. Observe that

$$|\max\{u(a^{I,n}, \theta), u(a^{II,n}, \theta)\}| \leq \phi(a_1^{I,n} - \theta_1)^2 + (a_2^{I,n} - \theta_2)^2.$$

For any converging sequence in $\mathbb{R}^2$, $a^n \rightarrow a$, the function

$$\phi(a_1^n - \theta_1)^2 + (a_2^n - \theta_2)^2 = (\phi\theta_1^2 + \theta_2^2) - 2(\phi\theta_1 a_1^n + \theta_2 a_2^n) + \phi(a_1^n)^2 + (a_2^n)^2$$

is dominated by an integrable function. This is the case, because the sequence (a^n) converges, hence it is bounded and $\phi(a_1^n)^2 + (a_2^n)^2 \leq M$ for some $M > 0$. Similarly, by the Cauchy-Schwarz inequality, $|\phi\theta_1 a_1^n + \theta_2 a_2^n| \leq \sqrt{M} \sqrt{\theta_1^2 + \theta_2^2}$. Therefore,

$$|\max\{u(a^{I,n}, \theta), u(a^{II,n}, \theta)\}| \leq \phi(a_1^{I,n} - \theta_1)^2 + (a_2^{I,n} - \theta_2)^2 \leq (\phi\theta_1^2 + \theta_2^2) + 2\sqrt{M} \sqrt{\theta_1^2 + \theta_2^2} + M,$$

for some $M > 0$. Because the variance of θ is finite, the dominating function is integrable.

It is also clear that

$$\max\{u(a^{I,n}, \theta), u(a^{II,n}, \theta)\} \rightarrow \max\{u(a^I, \theta), u(a^{II}, \theta)\}, \text{ for each } \theta.$$

Therefore by the dominated convergence theorem,

$$\int_{\Theta} \max\{u(a^{I,n}, \theta), u(a^{II,n}, \theta)\} dF \rightarrow \int_{\Theta} \max\{u(a^I, \theta), u(a^{II}, \theta)\} dF,$$

and the function $v(a^I, a^{II})$ is continuous.

As a second step, we establish that the following maximization problem has a solution:

$$\max_{a^I, a^{II} \in \mathbb{R}^2} v(a^I, a^{II}) \tag{A.6}$$

The function $v(a^I, a^{II})$ is bounded above by 0 and therefore a supremum exists, say $\bar{v}$. Moreover, setting $a^I = a^{II} = \mathbb{E}[\theta]$ guarantees a payoff of $-\phi \text{Var}[\theta_1] - \text{Var}[\theta_2]$ and therefore $\bar{v} \geq -\phi \text{Var}[\theta_1] - \text{Var}[\theta_2]$.

If $\bar{v} = -\phi \text{Var}[\theta_1] - \text{Var}[\theta_2]$, then the supremum is attained by $a^I = a^{II} = \mathbb{E}[\theta]$ and therefore a maximum exists.

Suppose instead that $\bar{v} > -\phi \text{Var}[\theta_1] - \text{Var}[\theta_2]$. Let $(a^{I,n}, a^{II,n})$ be a sequence such that $v(a^{I,n}, a^{II,n}) \rightarrow \bar{v}$. We want to show that the sequence $(a^{I,n}, a^{II,n})$ is bounded.

Suppose it is not. If $\|a^{k,n}\| \rightarrow \infty$, then $u(a^{k,n}, \theta) \rightarrow -\infty$ for each θ .

If $\|a^{k,n}\| \rightarrow \infty$ for both $k = I, II$, then $\max\{u(a^{I,n}, \theta), u(a^{II,n}, \theta)\} \rightarrow -\infty$ and therefore $v(a^{I,n}, a^{II,n}) \rightarrow -\infty$ and thus does not converge to $\bar{v}$.

If $\|a^{k,n}\| \rightarrow \infty$ for only one $k = I, II$, then $a^{-k,n}$ is bounded and admits a convergent subsequence to a^{-k} . Taking such subsequence, we get $\max\{u(a^{k,n}, \theta), u(a^{-k,n}, \theta)\} \rightarrow u(a^{-k}, \theta)$ for each θ . Using the dominated convergence theorem in a similar way as above, we get

$$v(a^{k,n}, a^{-k,n}) \rightarrow \int_{\Theta} u(a^{-k}, \theta) dF \leq -\phi \text{Var}[\theta_1] - \text{Var}[\theta_2].$$

But we assumed that the supremum $\bar{v} > -\phi \text{Var}[\theta_1] - \text{Var}[\theta_2]$, a contradiction.

Therefore, the sequence $(a^{I,n}, a^{II,n})$ is bounded and admits a convergent subsequence. By continuity, a maximum then exists.

To conclude the proof, note that the maximization problem (A.6) gives the Perfect Bayesian Equilibrium strategies of the common-interest game where the sender chooses a strategy $\mu : \Theta \rightarrow \{1, 2\}$ and the receiver chooses $(a^I, a^{II}) \in \mathbb{R}^2 \times \mathbb{R}^2$ to maximize

$$\max_{\mu, \alpha} \int_{\Theta} \mathbb{1}[\mu(\theta) = 1] u(a^I, \theta) + \mathbb{1}[\mu(\theta) = 2] u(a^{II}, \theta) dF. \quad (\text{A.7})$$

The receiver's beliefs are set according to Bayes' rule on path and any off path message is assigned a belief associated to an on path message. $\square$

Proposition 1 is a corollary of Lemma A1.

Proof of Proposition 1. Note first that $a^I = a^{II} = \mathbb{E}[\theta]$ is not a solution of the maximization problem (A.7), as any arbitrary partition of Θ and the best-reply to it would give strictly

higher payoffs. This means that the solution to (A.7) is a non-constant μ . Moreover, the strategy $\mu : \Theta \rightarrow \{1, 2\}$ trivially satisfies the IVP. Therefore, an equilibrium score exists. $\square$

C Proof of Remark 3

Fix $L > 0$ and $K \in \mathbb{N}$.

We prove Remark 3 assuming that the state space is $[-L, L]^2$. The proof can be straightforwardly extended to $\mathbb{R}^2$ by extending the scores we will define to the rest of $\mathbb{R}^2$, but at the expense of heavy notation that would obscure the argument.

We now construct a sequence of scores indexed by ϵ such that the sequence of associated payoffs converges to $v^* = 0$.

As an intermediate step, we build a sequence of correspondences. Figure 4 illustrates a correspondence in the sequence for $K = 3$. We define the correspondence $S_\epsilon(\theta)$ on $[-L, L]^2$ as follows,

For $i \in \{1, 2, \dots, K\}$ and $j \in \{1, 2, \dots, K - 1\}$, we have:

- $S_\epsilon(\theta) \ni K(j - 1) + i$ if

$$\theta \in [-L + (i - 1)\frac{2L}{K}, -L + i\frac{2L}{K}] \times [-L + (j - 1)\frac{2L}{K}, -L + j\frac{2L}{K} - \frac{\epsilon}{2^{i-1}}];$$

- for $m \in \{1, 2, \dots, K - 1\}$, $S_\epsilon(\theta) \ni K(j - 1) + i + m$, if

$$\theta \in [-L + (i - 1)\frac{2L}{K}, -L + i\frac{2L}{K}] \times [-L + j\frac{2L}{K} - \frac{\epsilon}{2^{i-1}}\frac{K - m}{K - 1}, -L + j\frac{2L}{K} - \frac{\epsilon}{2^{i-1}}\frac{K - m - 1}{K - 1}];$$

For $i \in \{1, 2, \dots, K\}$, we have $S_\epsilon(\theta) \ni K(K - 1) + i$ if

$$\theta \in [-L + (i - 1)\frac{2L}{K}, -L + i\frac{2L}{K}] \times [-L + (K - 1)\frac{2L}{K}, L].$$

Broadly speaking, the correspondence splits the state space in a $K \times K$ grid of cells. For all rows except the top one, cells are further subdivided in rectangular areas. The correspondence maps each state in the same area, including the boundaries of the area, to the same message in

7	8	9
6	7 6	7 8
5	5	6
4		
3	4 3	4 5
2	2	3
1		

Figure 4: Illustration of s_ϵ for $K = 3$.

$\{1, 2, \dots, K^2\}$ in a way that ensures that states in any two areas that share a border are mapped into messages that do not differ by more than 1. The correspondence maps any point in the interior of an area to a single message, and any point on a boundary to multiple messages. We define score s_ϵ by selecting appropriately the mapping from state to messages at the boundaries in S_ϵ .

Define $s_\epsilon(\theta) = \lfloor \text{med } S_\epsilon(\theta) \rfloor$ where med denotes the median. This defines the score on $[-L, L]^2$.

The score s_ϵ as defined is indeed a score since for any fixed θ_k , $s_\epsilon(\theta_{-k}, \theta_k)$ increases by at most one increment as θ_{-k} increases.

Let $v(s; K) = \int_{\Theta} u(\alpha(s(\theta)), \theta) dF$ s.t. $\alpha(m) = \mathbb{E}[\theta|m]$ for all $m \in s(\Theta)$.

As $\epsilon \rightarrow 0$, s_ϵ converges pointwise to s defined as follows:

Let $R(i, j; K) = [-L + (i-1)\frac{2L}{K}, -L + i\frac{2L}{K}] \times [-L + (j-1)\frac{2L}{K}, -L + j\frac{2L}{K}]$.

for $1 \leq i \leq K, 1 \leq j \leq K$, if $\theta \in R(i, j; K)$, $S(\theta) = K(j-1) + i$ and $s(\theta) = \lfloor \text{med } S(\theta) \rfloor$.

Note that $s(\theta)$ is not a score.

For each $m \in \{0, \dots, K^2\}$, let $A_\epsilon(m) = \{\theta : s_\epsilon(\theta) = m\}$ and $A(m) = \{\theta : s(\theta) = m\}$. Clearly, we have $\mathbb{1}[\theta \in A_\epsilon(m)] - \mathbb{1}[\theta \in A(m)] \rightarrow 0$ for all θ . Therefore, applying the dominated convergence theorem, we have

$$\alpha_\epsilon(m) := \mathbb{E}[\theta | A_\epsilon(m)] = \frac{\int \theta \mathbb{1}[\theta \in A_\epsilon(m)] dF}{\int \mathbb{1}[\theta \in A_\epsilon(m)] dF} \rightarrow \frac{\int \theta \mathbb{1}[\theta \in A(m)] dF}{\int \mathbb{1}[\theta \in A(m)] dF} = \mathbb{E}[\theta | A(m)] := \alpha(m).$$

Therefore, $v(s_\epsilon; K) = \sum_m \int \mathbb{1}[\theta \in A_\epsilon(m)] u(\alpha_\epsilon(m), \theta) dF \rightarrow \sum_m \int \mathbb{1}[\theta \in A] u(\alpha, \theta) dF = v(s; K)$.

We now argue that $v(s; K) \rightarrow 0$ as $K \rightarrow \infty$.

For each θ , $\sum_m \mathbb{1}[\theta \in A(m; K)] u(\mathbb{E}[\theta | \theta \in A(m; K)], \theta) \rightarrow 0$ as $K \rightarrow \infty$. Applying the dominated convergence theorem again, we have $v(s; K) \rightarrow 0$.

D Proof of Proposition 2

For a score s , let M be the image of the score. By the disintegration theorem (see e.g., Kallenberg (2021), Theorem 3.4), there exists a system of beliefs on M , $\{F(\cdot | m)\}_{m \in M} \subseteq \Delta(\Theta)$ such that $F(\cdot | m)$ is a regular conditional probability distribution and this system of beliefs is almost everywhere uniquely defined (with respect to the distribution of m induced by s and F). The score s is an equilibrium score if for some system of beliefs derived from s and F , $\alpha(m) = \mathbb{E}[\tilde{\theta} | s(\theta) = m]$ and s satisfies the IC constraints when the receiver's strategy is α .

For some receiver strategy α , denote by $\alpha(M)$ the image of $\alpha(\cdot)$ and let $\Theta(a) := \{\theta : \alpha(s(\theta)) = a\}$. For any two points, $x, y \in \mathbb{R}^2$, with a slight abuse of notation, let $[x, y] = \text{conv}\{x, y\}$, $(x, y) = [x, y] \setminus \{x, y\}$ and $[x, y) = [x, y] \setminus \{y\}$. Finally, let $\ell(x, y)$ be the line connecting the points x, y .

We also note that the utility function defines a norm:

$$\|\theta\|_\phi := \sqrt{\phi \theta_1^2 + \theta_2^2}.$$

When $\phi = 1$, this is the usual Euclidean norm. We have that

$$u(a, \theta) \geq u(a', \theta) \Leftrightarrow \|a - \theta\|_\phi \leq \|a' - \theta\|_\phi.$$

The following lemma will be used throughout the proof.

Lemma A2. *Let $a, a' \in \mathbb{R}^2$. If $u(a, \theta) \geq u(a', \theta)$, then $u(a, \theta') > u(a', \theta')$ for all $\theta' \in [a, \theta]$.*

Proof. Take $\theta' \in [a, \theta]$. First assume that $a' \notin \ell(a, \theta')$. Note that

$$\|a - \theta\|_\phi \leq \|a' - \theta\|_\phi < \|\theta' - \theta\|_\phi + \|a' - \theta'\|_\phi, \quad (\text{A.8})$$

where the last inequality holds by the triangle inequality and is strict because θ, θ' and a' are not collinear. Note also that

$$\begin{aligned} \|\theta' - \theta\|_\phi + \|a - \theta'\|_\phi &= \|a - \theta\|_\phi < \|\theta' - \theta\|_\phi + \|a' - \theta'\|_\phi \\ \Rightarrow u(a, \theta') &> u(a', \theta'), \end{aligned}$$

where the equality holds as a, θ and θ' are collinear, and the first inequality follows from (A.8).

If instead $a' \in \ell(a, \theta')$, we must have $a' \notin (a, \theta]$, otherwise $u(a, \theta) < u(a', \theta)$. But then, either $a \in (a', \theta')$ or $\theta \in (\theta', a')$. In both cases, $u(a, \theta') > u(a', \theta')$. $\square$

Let s be an equilibrium score with associated receiver strategy α . The next lemmas characterize s as the characteristics of α vary.

We first consider the case in which all points in $\alpha(M)$ are isolated.

Lemma A3. *If all points in $\alpha(M)$ are isolated, then s is equivalent to a coarsely linear score.*

Proof. For any two $a, a' \in \alpha(M)$, let $\Theta^\geq(a, a') := \{\theta : u(a, \theta) \geq u(a', \theta)\}$. This set is a half-space:

$$u(a, \theta) \geq u(a', \theta) \Leftrightarrow -2\theta_1 a_1 \phi + a_1^2 \phi - 2\theta_2 a_2 + a_2^2 \geq -2\theta_1 a'_1 \phi + a_1'^2 \phi - 2\theta_2 a'_2 + a_2'^2.$$

Similarly, let $\Theta^=(a, a') := \{\theta : u(a, \theta) = u(a', \theta)\}$. This set is a line.

If $|\alpha(M)| = 2$, the set $\Theta^=(a, a')$ determines the half-space defining a coarsely linear score.

Suppose there are three points $a^1, a^2, a^3 \in \alpha(M)$ and $m^i \in \alpha^{-1}(a^i)$ for $i = 1, 2, 3$ such that (i) $m^1 < m^2 < m^3$ and (ii) for any action $a' \in \alpha(M) \setminus \{a^1, a^2, a^3\}$, every $m \in \alpha^{-1}(a')$ satisfies $m > m^3$ or $m < m^1$.

Suppose that $\Theta^=(a^1, a^2)$ and $\Theta^=(a^2, a^3)$ are not parallel. Then $\Theta(a^2) \subseteq \Theta^{\geq}(a^2, a^1) \cap \Theta^{\geq}(a^2, a^3)$ and the set $\Theta^{\geq}(a^2, a^1) \cap \Theta^{\geq}(a^2, a^3)$ is a polyhedron with an extreme point at $\Theta^=(a^2, a^1) \cap \Theta^=(a^2, a^3)$.

Clearly $\{a^1, a^3\} \cap \Theta^{\geq}(a^2, a^1) \cap \Theta^{\geq}(a^2, a^3) = \emptyset$. Moreover, we can draw a curve from a^1 to a^3 in $\Theta \setminus (\Theta^{\geq}(a^2, a^1) \cap \Theta^{\geq}(a^2, a^3))$. By the IVP, there must be θ' on that curve such that $s(\theta') = m^2$, a contradiction. $\square$

We consider next the case in which not all points in $\alpha(M)$ are isolated.

Lemma A4. *Let a be a limit point in $\alpha(M)$. Then $\text{int } \Theta(a) = \emptyset$.*

Proof. We proceed by contradiction. Suppose $\text{int } \Theta(a) \neq \emptyset$. Let $\Theta^>(a) = \{\theta : u(a, \theta) > u(a', \theta), \text{ for all } a' \in \alpha(M) \setminus \{a\}\}$ and $\Theta^=(a) = \{\theta \in \Theta(a) : u(a, \theta) = u(a', \theta), \text{ for some } a' \in \alpha(M) \setminus \{a\}\}$. We have $\Theta^>(a) \subseteq \Theta(a) \subseteq \Theta^=(a) \cup \Theta^>(a)$.

The set $\Theta^=(a)$ has an empty interior as by Lemma A2, if $u(a, \theta) = u(a', \theta)$, $u(a, \theta) < u(a', \theta)$ for all $\theta' \in [a', \theta)$. Therefore, $\Theta^>(a)$ has a non-empty interior.

We also have that $\Theta^=(a) \subseteq \text{boundary } \Theta^>(a)$ as by Lemma A2, if $u(a, \theta) \geq u(a', \theta)$, $u(a, \theta) > u(a', \theta)$ for all $\theta' \in [a, \theta)$. Therefore, for any $\theta \in \Theta^=(a)$ and open ball $B_\epsilon(\theta)$, $B_\epsilon(\theta)$ intersects with both $\Theta^>(a)$ and $\Theta \setminus \Theta^>(a)$.

Next, we argue that $\Theta^>(a)$ is convex. Observe that

$$u(\theta, a) > u(\theta, a') \Leftrightarrow -2\theta_1 a_1 \phi + a_1^2 \phi - 2\theta_2 a_2 + a_2^2 > -2\theta_1 a'_1 \phi + a_1'^2 \phi - 2\theta_2 a'_2 + a_2'^2.$$

The inequality is preserved under convex combinations of θ , so $\Theta^>(a)$ is convex.

If the interior of $\Theta^>(a)$ is not empty and $\Theta^>(a)$ is convex, then the boundary of $\Theta^>(a)$ has measure zero in $\mathbb{R}^2$ (see e.g., Lang, 1986). This also implies that $\Theta^=(a)$ has measure zero.

Moreover, since $\mathbb{E}[\theta|s(\theta) = m] = a$ for all $m \in \{m' \in M : \alpha(m') = a\}$, we have

$$\mathbb{E}[\theta|\theta \in \Theta(a)] = a.$$

Therefore,

$$\mathbb{E}[\theta|\theta \in \Theta(a)] = \mathbb{E}[\theta|\theta \in \Theta^>(a)] = \mathbb{E}[\theta|\theta \in \text{int } \Theta^>(a)] = a,$$

which implies $a \in \text{int } \Theta^>(a)$ by the full support assumption. But then, because a is a limit point of $\alpha(M)$, it means that $\text{int } \Theta^>(a)$ intersects with $\alpha(M)$ at a point different than a , i.e., there is a point $a' \in \alpha(M) \cap \text{int } \Theta^>(a)$, $a' \neq a$ and associated message m' with $\alpha(m') = a'$ such that $u(a, a') < u(\alpha(m'), a') = 0$. Therefore, type $\theta = a'$ has a profitable deviation, a contradiction. Hence, $\text{int } \Theta(a) = \emptyset$. $\square$

Let A_I be the set of isolated points in $\alpha(M)$ and A_L be the set of limit points in $\alpha(M)$.

Lemma A5. *For any limit point of $\alpha(M)$, a , $\Theta(a) = \ell(\theta, \theta')$ for some θ and θ' and all lines are parallel.*

Proof. Take $a \in \alpha(M \setminus \{\inf M, \sup M\})$.

First, we show that there are θ and θ' such that $\Theta(a) \subseteq \ell(\theta, \theta')$.

From the proof of Remark 2, $|\Theta(a)| > 1$ and therefore $\Theta(a) \neq \{a\}$.

We first show that for any two $\theta, \theta' \in \Theta(a)$, $a \in \ell(\theta, \theta')$. Suppose it is not the case. By Lemma A2, we can assume that for $\theta^\dagger \in \{\theta, \theta'\}$ we have $u(\theta^\dagger, a) > u(\theta^\dagger, a')$ for all $a' \in \alpha(M) \setminus \{a\}$. Otherwise, we can just take another point in $(a, \theta^\dagger)$. As argued in the proof of Lemma A4, for any $\theta'' \in \text{conv}\{\theta, \theta', a\}$, we have $u(a, \theta'') > u(a', \theta'')$ for all $a' \in \alpha(M) \setminus \{a\}$ and thus $\theta'' \in \Theta(a)$. Since θ, θ', a are not aligned, $\text{conv}\{\theta, \theta', a\}$ has a non-empty interior. Therefore, $\text{int } \Theta(a)$ has a non-empty interior. A contradiction.

To prove that $\Theta(a) = \ell(\theta, \theta')$ if a is a limit point and $a \neq \sup M, \inf M$, it is then enough to show that the set $\Theta(a)$ is unbounded in both directions. To see this, take some $\theta \in \Theta(a)$ and let $m = s(\theta)$. We can repeat the same argument as in Remark 2. Let m_1 and m_2 satisfy $m_1 < m < m_2$ and pick θ^1 and θ^2 such that $s(\theta^1) = m_1$ and $s(\theta^2) = m_2$.

If $\Theta(a)$ is bounded in one direction, we can find a curve such that this curve does not intersect with $\Theta(a)$. By the IVP, there must be θ' on that curve such that $s(\theta') = m$ and therefore $\theta' \in \Theta(a)$, a contradiction. Therefore, $\Theta(a) = \ell(\theta, \theta')$.

Let a and a' be limit points of $\alpha(M \setminus \{\inf M, \sup M\})$ such that $a \neq a'$. Because $\Theta(a) \cap \Theta(a') = \emptyset$, the lines $\Theta(a)$ and $\Theta(a')$ must be parallel.

Denote by $\ell_s(a)$ the line that goes through a and has the same slope as $\Theta(a')$ for some $a' \in A_L \setminus \{\inf M, \sup M\}$.

To conclude the proof, we will show that if $\inf M \in M$, then $\alpha(\inf M)$ is not a limit point. (The proof for the sup follows the same lines). Suppose it is not the case and that $m = \min M$ exists and $\alpha(m)$ is a limit point of $\alpha(M)$. Therefore, there is a neighborhood of $\alpha(m)$, denote it O , such that for all $\theta \in O$, $\sup_{a \in A_L} u(a, \theta) > \sup_{a \in A_I \cup \{\alpha(\sup M)\}} u(a, \theta)$ and for all $a \in O \cap \alpha(M)$, $a \in A_L$.¹⁵ That is, types in O are closer to points in A_L than to points in A_I and are in A_L and there are no isolated points in O .

Take a point in $\theta \in \ell_s(\alpha(m)) \cap O$. It cannot be that $\alpha(s(\theta)) \in A_I \cup \{\alpha(\sup M)\}$ by definition of O . It also cannot be that $\alpha(s(\theta)) \in A_L \setminus \{\alpha(m)\}$ as $\theta \in \ell_s(\alpha(m))$ and therefore not in $\ell_s(a)$ for $a \in A_L \setminus \{\alpha(m), \alpha(\sup M)\}$. Therefore, $\alpha(s(\theta)) = \alpha(m)$ and there is more than one point in $\Theta(\alpha(m))$. By a similar argument as above, it must be that $\Theta(\alpha(m)) \subseteq \ell_s(\alpha(m))$.

Let Θ^+ and Θ^- denote the two open half-spaces defined by the line $\ell_s(\alpha(m))$. Suppose there are $a^+ \in \Theta^+$ and $a^- \in \Theta^-$ such that $a^+, a^- \in O \cap \alpha(M)$, i.e., there are actions played in equilibrium in A_L that are on both sides of $\ell_s(\alpha(m))$. Note that $\ell_s(a^-) \subset \Theta^-$.

Suppose without loss of generality that $m^+ = s(a^+) > m^- = s(a^-)$. By definition, $m^- > m$. Take two points $\theta^+ \in \ell_s(a^+)$, $\theta^m \in \Theta(\alpha(m))$. We can draw a curve between θ^+ and θ^m that is entirely in Θ^+ (except at θ^m). By IVP, there must be θ' on that curve such that $s(\theta') = m^-$. But $\theta' \in \Theta^+$ and $\notin \ell_s(a^-) = \Theta(a^-)$, a contradiction.

Therefore all $\theta \in O \cap \alpha(M)$ are in the same half-space, say Θ^- . But types in $\Theta^+ \cap O$ should prefer sending messages that induce $a \in A_L$, contradicting that $\Theta(a) \subseteq \ell_s(a)$. $\square$

As defined in the proof of Lemma A5, denote by $\ell_s(a)$ the line that goes through a and has the same slope as $\Theta(a')$ for some $a' \in A_L$.

Lemma A6. *If there are some limit points in $\alpha(M)$, then all points in $\alpha(M)$ are limit points.*

Proof. Let $\Theta^\dagger = \cup_{a \in \text{cl} A_L} \Theta(a) = \cup_{a \in \text{cl} A_L} \ell_s(a)$.

¹⁵Using the convention that if $\sup M \notin M$, then $\{\alpha(\sup M)\} = \emptyset$.

Take $a \in \arg \max_{a' \in A_I} \sup_{\theta \in \Theta^\dagger} u(a', \theta)$ and $\theta^\dagger \in \arg \max_{\theta \in \Theta^\dagger} u(a, \theta)$. The points a and $\theta^\dagger$ are the two points in A_I and $\Theta^\dagger$ with minimal (weighted) distance between the two. Moreover, this distance is bounded away from zero either by the definition of isolated points if $\theta^\dagger \in A_L$ or by the optimality of generating an action in A_L for states arbitrarily close to $\theta^\dagger$ if $\theta^\dagger \notin A_L$.

Note that $\theta^\dagger$ is on the boundary of $\Theta^\dagger$, otherwise there is another point in $\Theta^\dagger$ closer to a . Take $\tilde{a} \in \text{cl}A_L$ such that $\theta^\dagger \in \ell_s(\tilde{a})$. Because the $\Theta^\dagger$ is a union of lines, if $\theta^\dagger \in \ell_s(\tilde{a})$ is on the boundary of $\Theta^\dagger$, then $\ell_s(\tilde{a})$ is on the boundary of $\Theta^\dagger$. We can therefore find a sequence $\theta^n \notin \Theta^\dagger$ with $\theta^n \rightarrow \tilde{a}$. By definition of isolated points, there is $\epsilon > 0$ such that $u(a, \tilde{a}) < -\epsilon$. But then for n large enough, θ^n prefers to induce an action in A_L , contradicting that $\theta^n \notin \Theta^\dagger$. $\square$

Proof of Proposition 2. All equilibrium scores are equivalent to linear or coarsely linear scores. Take an equilibrium score s with associated receiver strategy α . Suppose all points in $\alpha(M)$ are isolated. Then by Lemma A3, the score is equivalent to a coarsely linear score.

Suppose some points in $\alpha(M)$ are limit point. Then by Lemma A6, all points in $\alpha(M)$ are limit points. By Lemma A5, the score is equivalent to a linear score.

All equilibrium payoffs are uniformly below commitment payoffs. To present a simpler proof of the second part of Proposition 2, we use Proposition 3 and 4. To prove these results we only use the first part of Proposition 2.

Given Proposition 4, any coarsely linear equilibrium payoffs $s_{\beta, C}$ is dominated in terms of payoffs by a linear score s_β with $\alpha(m) = \left(\frac{\beta_1}{\beta_1^2 + \phi\beta_2^2}m, \frac{\phi\beta_2}{\beta_1^2 + \phi\beta_2^2}m \right)$.

The payoffs from any score β with receiver strategy $\alpha(m) = \left(\frac{\beta_1}{\beta_1^2 + \phi\beta_2^2}m, \frac{\phi\beta_2}{\beta_1^2 + \phi\beta_2^2}m \right)$ is

$$\begin{aligned} & \mathbb{E}[-\phi(\theta_1 - \frac{\beta_1}{\beta_1^2 + \phi\beta_2^2}(\beta_1\theta_1 + \beta_2\theta_2))^2 - (\theta_2 - \frac{\phi\beta_2}{\beta_1^2 + \phi\beta_2^2}(\beta_1\theta_1 + \beta_2\theta_2))^2] \\ &= -\mathbb{E}[\frac{(\phi\beta_2\theta_1 - \beta_1\theta_2)^2}{\beta_1^2 + \phi\beta_2^2}] \\ &= -\frac{1}{\beta_1^2 + \phi\beta_2^2}(\phi^2\beta_2^2\text{Var}[\theta_1] + \beta_1^2\text{Var}[\theta_2] - 2\phi\beta_1\beta_2\text{Cov}[\theta_1, \theta_2]). \end{aligned}$$

This expression is uniformly below 0 across all β_1, β_2 : if $\beta_i = 0$, then the loss is proportional to $\text{Var}[\theta_{-i}]$.

If $\beta_i \neq 0$ for $i = 1, 2$, then letting $x = \beta_1/\beta_2$, we can rewrite the loss as

$$\frac{\phi^2 \text{Var}[\theta_2] + x^2 - 2\phi x \text{Cov}[\theta_1, \theta_2]}{x^2 + \phi}.$$

It is easy to verify that this expression is bounded away from zero by minimizing with respect to x . $\square$

E Proofs of Proposition 3 and Proposition 4

The payoff function u yields a well-defined inner product and norm:

$$\begin{aligned} \langle \theta, \theta' \rangle_\phi &= \phi \theta_1 \theta'_1 + \theta_2 \theta'_2, \\ ||\theta||_\phi^2 &= \phi \theta_1^2 + \theta_2^2, \text{ for all } \theta \in \mathbb{R}^2. \end{aligned}$$

Any line ℓ in $\mathbb{R}^2$ can be written as

$$\ell = \{\theta_0 + \lambda x : \lambda \in \mathbb{R}\},$$

where $x \in \mathbb{R}^2 \setminus \{0\}$ is a *direction vector* of the line ℓ .

Two lines ℓ, ℓ' are *parallel* if for two direction vectors x, x' of ℓ, ℓ' , there is $c \neq 0$ such that $x = c x'$.

Two lines are *orthogonal* (with respect to the inner product $\langle \cdot, \cdot \rangle_\phi$) if for two direction vectors x, x' of ℓ, ℓ' , $\langle x, x' \rangle_\phi = 0$.

Moreover, let E_β and $E_{\beta,C}$ denote, respectively, the images of $\mathbb{E}[\tilde{\theta}|s_\beta(\theta) = m]$ and $\mathbb{E}[\tilde{\theta}|s_{\beta,C}(\theta) = m]$:

$$\begin{aligned} E_\beta &:= \{\mathbb{E}[\tilde{\theta}|s_\beta(\theta) = m] : m \in \mathbb{R}\}. \\ E_{\beta,C} &:= \{\mathbb{E}[\tilde{\theta}|s_{\beta,C}(\theta) = m] : m \in M\}. \end{aligned}$$

Lemma A7. *Let $\beta \in \mathbb{R}^2 \setminus \{0\}$. The following are equivalent*

1. *The score s_β is an equilibrium score.*

2. The set E_β is a line and is orthogonal to $s_\beta^{-1}(m)$ for every m .

Proof. (2) $\Rightarrow$ (1). Pick a $\theta^* \in \mathbb{R}^2$ and let $m^* = \beta_1 \theta_1^* + \beta_2 \theta_2^*$. Also, let $m \neq m^*$. We show that a deviation to message m in state θ^* is unprofitable.

Since E_β is orthogonal to $s_\beta^{-1}(m^*)$ and $\theta^*, \mathbb{E}[\theta|m^*] \in s_\beta^{-1}(m^*)$, for any $\mathbb{E}[\theta|m] \in E_\beta \setminus \{\mathbb{E}[\theta|m^*]\}$, by Pythagorean theorem,

$$\|\theta^* - \mathbb{E}[\theta|m]\|_\phi = \|\theta^* - \mathbb{E}[\theta|m^*]\|_\phi + \|\mathbb{E}[\theta|m] - \mathbb{E}[\theta|m^*]\|_\phi.$$

Since

$$\|\theta^* - \mathbb{E}[\theta|m^*]\|_\phi + \|\mathbb{E}[\theta|m] - \mathbb{E}[\theta|m^*]\|_\phi > \|\theta^* - \mathbb{E}[\theta|m^*]\|_\phi,$$

by transitivity:

$$\|\theta^* - \mathbb{E}[\theta|m]\|_\phi > \|\theta^* - \mathbb{E}[\theta|m^*]\|_\phi.$$

Hence indeed in state θ^* the sender has no profitable deviation to any message $m \neq m^*$. Hence the linear score associated with β is an equilibrium score.

(1) $\Rightarrow$ (2). We proceed by contradiction. Suppose that E_β is not a line or not orthogonal to $s_\beta^{-1}(m^*)$ for some m^* .

Therefore, there exist $m_A, m_B \in \mathbb{R}$ with $m_A \neq m_B$ such that the line ℓ_{AB} through $\mathbb{E}[\theta|m_A] \in E_\beta$ and $\mathbb{E}[\theta|m_B] \in E_\beta$ is not orthogonal to $s_\beta^{-1}(m_A)$.

Let $\ell_{\perp,B}$ be the line orthogonal to ℓ_{AB} through $e_\beta(m_B)$. The line $\ell_{\perp,B}$ cannot be both orthogonal to ℓ_{AB} and parallel to $s_\beta^{-1}(m_A)$.¹⁶ Therefore, $\ell_{\perp,B}$ and $s_\beta^{-1}(m_A)$ must intersect and there exists a $\theta^* \in s_\beta^{-1}(m_A) \cap \ell_{\perp,B}$. By an argument akin to the one in the if direction of our proof:

$$\|\theta^* - \mathbb{E}[\theta|m_B]\|_\phi < \|\theta^* - \mathbb{E}[\theta|m_A]\|_\phi.$$

Hence the sender has a profitable deviation from $m = m_A$ to $m = m_B$ in state θ^* . This is a contradiction since $\theta^* \in s_\beta^{-1}(m_A)$. $\square$

Lemma A8. Let $\beta \in \mathbb{R}^2 \setminus \{0\}$. The following are equivalent

¹⁶If both hold then, for some $c \neq 0$ and $u_\perp, u_{AB}, u_m$ some direction vectors of $\ell_{\perp,B}, \ell_{AB}, s_\beta^{-1}(m_A)$, we have $0 = \langle u_\perp, u_{AB} \rangle_\phi = \langle cu_m, u_{AB} \rangle_\phi = c \langle u_m, u_{AB} \rangle_\phi \neq 0$ where the first equality holds because $\ell_{\perp,B} \perp \ell_{AB}$, the second because $\ell_{AB}, s_\beta^{-1}(m_A)$ are parallel and the inequality because $\ell_{AB}, s_\beta^{-1}(m_A)$ are not orthogonal.

1. The set E_β is a line and is orthogonal to $s_\beta^{-1}(m)$ for every m .

2. For every m ,

$$\mathbb{E}[\theta | s_\beta(\theta) = m] = \left(\frac{\beta_1}{\beta_1^2 + \phi\beta_2^2}m, \frac{\phi\beta_2}{\beta_1^2 + \phi\beta_2^2}m \right).$$

Proof. (2) $\Rightarrow$ (1). We verify that the line generated by E_β is orthogonal to $s_\beta^{-1}(m)$ for every m . The line E_β generated by $\mathbb{E}[\theta | s_\beta(\theta) = m]$ corresponds to

$$\left\{ \frac{m}{\beta_1^2 + \phi\beta_2^2} u_e : m \in \mathbb{R} \right\},$$

where $u_e := (\beta_1, \phi\beta_2)$. For every m , the line $s_\beta^{-1}(m)$ can be written as:

$$\left\{ \left(0, \frac{m}{\beta_2}\right) + \theta_1 u_m : \theta_1 \in \mathbb{R} \right\},$$

where $u_m := (1, -\frac{\beta_1}{\beta_2})$. The direction vectors u_e and u_m satisfy:

$$\langle u_e, u_m \rangle_\phi = \phi\beta_1 + \phi\beta_2 \frac{-\beta_1}{\beta_2} = 0,$$

hence the line E_β is orthogonal to $s_\beta^{-1}(m)$ for every m .

(1) $\Rightarrow$ (2). If E_β is a line then we can write as

$$E_\beta = \{(b_1, b_2) + mu : m \in \mathbb{R}\}.$$

The lines $E_\beta, s_\beta^{-1}(m)$ are orthogonal for each m if

$$\langle u, u_m \rangle_\phi = 0 \Leftrightarrow \phi u_1 + u_2 \frac{-\beta_1}{\beta_2} = 0.$$

Without loss of generality, we can set $u_2 = \frac{\phi\beta_2}{\beta_1^2 + \phi\beta_2^2}$ as two multiples of direction vectors generate the same line. Therefore, we get $u_1 = \frac{\beta_1}{\beta_1^2 + \phi\beta_2^2}$. Moreover, $\mathbb{E}[\theta_1] = 0$ and $E[m] = \beta_1 \mathbb{E}[\theta_1] + \beta_2 \mathbb{E}[\theta_2] = 0$. By the law of iterated expectation,

$$\mathbb{E}_m[\mathbb{E}[\theta_i | m]] = \mathbb{E}_m\left[\frac{\beta_1^2}{\beta_1^2 + \phi\beta_2^2}m + b_1\right] = 0.$$

Therefore, $b_1 = 0$. By a similar reasoning, $b_2 = 0$.

□

Proof of Proposition 3. The result is an immediate consequence of Lemma A7 and Lemma A8.

□

Proof of Proposition 4. The proof of Proposition 4 follows the same lines as the proof of Proposition 3 replacing the sets $s_\beta^{-1}(m)$ with the sets of types indifferent between m and $m - 1$.

□

F Proof of Proposition 5

For any strategy $s(\theta) = \beta'\theta$, we have the unconditional distribution over messages m induced by the score s , $m \sim N(0, \sigma_s^2)$ where $\sigma_s^2 = \beta_1^2\sigma_1^2 + \beta_2^2\sigma_2^2 + 2\beta_1\beta_2\rho\sigma_1\sigma_2 = \beta'\Sigma\beta$. We also have that $\text{Cov}(\theta_i, m) = \sigma_{is} = \beta_i\sigma_i^2 + \beta_j\rho\sigma_1\sigma_2$. Therefore, $(\sigma_{1s}, \sigma_{2s})' = \Sigma\beta$.

The payoff of the sender can be rewritten, up to a constant, as

$$-a'\Phi a + 2a'\Phi\theta.$$

Therefore, the ex-ante payoff – given that the best-reply to m is $\alpha(m) = \frac{\Sigma\beta}{\beta'\Sigma\beta}m$ – is

$$\begin{aligned} & \mathbb{E}[-\alpha(m)'\Phi\alpha(m) + 2\alpha(m)'\Phi\theta] \\ &= \mathbb{E}\left[-\frac{\beta'\Sigma}{\beta'\Sigma\beta}m\Phi\frac{\Sigma\beta}{\beta'\Sigma\beta}m + 2\frac{\beta'\Sigma}{\beta'\Sigma\beta}m\Phi\theta\right] \\ &= \frac{\beta'\Sigma\Phi\Sigma\beta}{\beta'\Sigma\beta}, \end{aligned}$$

where the last equality follows from $\mathbb{E}[m^2] = \beta'\Sigma\beta$ and $\mathbb{E}[\theta m] = \Sigma\beta$. The matrix $\Sigma\Phi\Sigma$ is positive semidefinite and symmetric. Therefore, $\frac{\beta'\Sigma\Phi\Sigma\beta}{\beta'\Sigma\beta}$ is a generalized Rayleigh quotient (see e.g., Parlett, 1998, Chapter 15) and the two stationary points, up to a rescaling of β , of $\frac{\beta'\Sigma\Phi\Sigma\beta}{\beta'\Sigma\beta}$ are the eigenvectors of $\Sigma^{-1}(\Sigma\Phi\Sigma) = \Phi\Sigma$, i.e., the points β such that there is $\lambda \in \mathbb{R}$ such that $\Phi\Sigma\beta = \lambda\beta$. Moreover, as generalized Rayleigh quotients attain a maximum and a minimum, one of the stationary points must correspond to a maximizer, the other to a minimizer.¹⁷

¹⁷If the state had more than two dimensions, there would be more stationary points/eigenvectors; yet, it

The equilibrium problem can be expressed as follows. Given a belief that the sender uses a linear strategy β , the receiver chooses $\alpha(m) = \frac{\Sigma\beta}{\beta'\Sigma\beta}m$. In equilibrium, the sender chooses a signal m for each realization of θ :

$$\max_m -\frac{\beta'\Sigma m\Phi\Sigma\beta m}{(\beta'\Sigma\beta)^2} + 2\frac{\beta'\Sigma m\Phi\theta}{\beta'\Sigma\beta}.$$

The objective function is quadratic in m and therefore the maximizer must satisfy the first-order condition:

$$m = \beta'\Sigma\beta \frac{\beta'\Sigma\Phi}{\beta'\Sigma\Phi\Sigma\beta}\theta.$$

Therefore, any equilibrium strategy must satisfy

$$\beta' = \beta'\Sigma\beta \frac{\beta'\Sigma\Phi}{\beta'\Sigma\Phi\Sigma\beta} \Leftrightarrow \beta = \frac{\beta'\Sigma\beta}{\beta'\Sigma\Phi\Sigma\beta}\Phi\Sigma\beta.$$

Take any equilibrium strategy β . From the equilibrium condition, β is an eigenvector of $\Phi\Sigma$ with eigenvalue $\frac{\beta'\Sigma\Phi\Sigma\beta}{\beta'\Sigma\beta}$.

Conversely, take an eigenvector β of $\Phi\Sigma$, with eigenvalue λ . Plugging in the equilibrium condition, we get

$$\beta = \beta'\Sigma\beta \frac{\Phi\Sigma\beta}{\beta'\Sigma\Phi\Sigma\beta} \Leftrightarrow \beta = \frac{\beta'\Sigma\beta}{\lambda\beta'\Sigma\beta}\lambda\beta, \quad (\text{A.9})$$

where the equivalence follows from $\Phi\Sigma\beta = \lambda\beta$ and $\beta'\Sigma\Phi = \lambda\beta'$. Equation (A.9) is satisfied and therefore β is an equilibrium strategy.

G Value of Ignorance

We show here, by way of an example, that sender and receiver *can* be better off if the sender is less informed.

Let $\phi = 1$, $\Theta = \{0, 1, 2\} \times \{0, 2\}$, $f(0, 0) = f(1, 2) = f(2, 2) = \frac{\epsilon}{3}$ and $f(0, 2) = f(1, 0) = f(2, 0) = \frac{1-\epsilon}{3}$, for some $\epsilon < \frac{5}{24}$. As in Section 3, we use the “graph-theoretic” definition of IVP. To apply it, we view $\{0, 1, 2\} \times \{0, 2\}$ as a simple graph where each $\theta \in \Theta$ is a vertex and the edges link adjacent points, i.e., an edge links two vertices θ, θ' such that either

would still be the case that one of the eigenvectors corresponds to a maximizer of the Rayleigh quotient, another to the minimizer.

$$|\theta_1 - \theta'_1| = 1 \text{ and } |\theta_2 - \theta'_2| = 0 \text{ or } |\theta_1 - \theta'_1| = 0 \text{ and } |\theta_2 - \theta'_2| = 2.$$

We consider the standard setting and a setting in which the sender does not observe θ_2 and therefore can only select strategies that assign the same message to any two states with the same θ_1 . In Figure 5, empty circles denote low-probability states and filled circles high-probability ones. The figure illustrates a score that assigns to each state (θ_1, θ_2) a message equal to θ_1 . We refer to this as score s_1 .

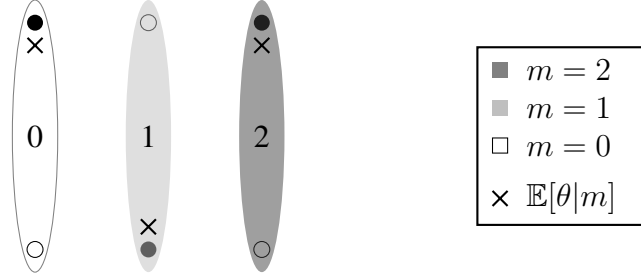


Figure 5

If the sender does not observe θ_2 , then score s_1 is an equilibrium strategy.¹⁸ We show next that any equilibrium score in the standard setting is associated with a larger loss than s_1 .

We proceed in two steps. First, we establish that any score such that $s(\theta) = s(\theta')$ for two high-probability states is associated with a larger loss than s_1 .

The expected loss associated with s_1 is: $4\epsilon(1 - \epsilon)$. Consider a score s such that $s(0, 2) = s(1, 0) = m'$. In any equilibrium in which $\mu(\theta) = s(\theta)$ for all θ , the loss conditional on $\theta \in \{(0, 2), (1, 0)\}$ is minimized if $\alpha(m') = (0.5, 1)$. For such action, the loss conditional on $\theta \in \{(0, 2), (1, 0)\}$ is $\frac{5}{4}$. Thus

$$Pr(\theta \in \{(0, 2), (1, 0)\}) \times \frac{5}{4}$$

is a lower bound on the loss from any score pooling $(0, 2)$ and $(1, 0)$ together. Note that

$$Pr(\theta \in \{(0, 2), (1, 0)\}) \times \frac{5}{4} = (1 - \epsilon) \frac{5}{6} > 4\epsilon(1 - \epsilon) \Leftrightarrow \epsilon < \frac{5}{24}.$$

Hence any score s such that $s(0, 2) = s(1, 0)$ is associated with a larger loss than s_1 . Similar

¹⁸For this strategy, $\mathbb{E}(\theta|m=0) = (0, 2(1 - \epsilon))$, and $\mathbb{E}(\theta|m=1) = (1, 2\epsilon)$. The expected loss, conditional on $\theta = (0, 0)$, is $4\epsilon(1 - \epsilon)$. Deviating to report $m = 1$ upon observing $\theta = (0, 0)$ induces a loss, conditional on $\theta = (0, 0)$, equal to $1 + 4(1 - \epsilon)(1 - 2\epsilon)^2$. As $1 + 4(1 - \epsilon)(1 - 2\epsilon)^2 > 4\epsilon(1 - \epsilon)$, the deviation is not profitable. Similar arguments show that the sender does not have any profitable deviation.

arguments apply for any score such that $s(\theta) = s(\theta')$ for any two high-probability states.

The second step amounts to showing that any score such that $s(0, 2) \neq s(1, 0) \neq s(2, 2)$ is not an equilibrium score.

To see this, let score s satisfy $s(0, 2) \neq s(1, 0) \neq s(2, 2)$. The IVP requires $s(1, 2) = 1$. We consider two cases: $s(1, 0) = 1$ and $s(1, 0) \neq 1$.

If $s(1, 0) = 1$, let, without loss, $s(0, 2) = 0$ and therefore $s(2, 2) = 2$. In this case, the IVP requires $s(0, 0) \neq 2$ and $s(2, 0) \neq 0$. This in turn implies that - in any equilibrium such that $\mu(\theta) = s(\theta)$ for all θ - the receiver chooses $\alpha(0) = (0, x)$, where $x \geq 2(1 - \epsilon)$ and $\alpha(1) = (y, z)$, where $z \leq 2\epsilon$. For any value of x, y and z , in state $(1, 2)$ the sender has a profitable deviation to report $m = 0$ instead of $m = 1$. So the score is not an equilibrium score.

If instead $s(1, 0) \neq 1$, let, without loss, $s(1, 0) = 2$, $s(0, 2) = 0$ and $s(2, 2) = 1$. The IVP requires $s(0, 0) = s(1, 2) = 1$ and $s(2, 0) \in \{1, 2\}$. Regardless of whether $s(2, 0) = 1$ or $s(2, 0) = 2$, in any equilibrium such that $\mu(\theta) = s(\theta)$ for all θ , in state $(0, 0)$ - the sender has a profitable deviation to report $m = 0$ instead of $m = 1$.

We conclude that any equilibrium score in the standard setting is associated with a larger loss than s_1 : in this example, sender and receiver can be better off if the sender does not observe the realization of θ_2 .